

Jack Ruan

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EXPERIENCE

Capital One | Software Engineer Feb. 2024 - *Present*

- Spearheaded the design and delivery of an extensible Claude Code AI agent after identifying support toil opportunities, engineering signal extraction and category classification over 3+ years of telemetry; established a modular sub-skills architecture that enabled cross-team contributions and automated triage to reduce MTTR.
- Drove enterprise Post-Quantum Cryptography (PQC) readiness by architecting a TLS handshake discovery engine to map protocol distribution, authoring technical specs, and partnering with AWS engineering to deliver support for NIST PQC algorithms in AL2023 2023.10 across 90+ Golden Docker Images and 22 Lambda runtimes.
- Automated Public Key Infrastructure (PKI) certificate lifecycle discovery across 1,000+ AWS accounts by architecting an event-driven serverless pipeline (Lambda, SQS, DynamoDB), eliminating 50% of manual operational toil and instrumenting 30+ services with OpenTelemetry distributed tracing.
- Eliminated critical audit compliance gaps across 3 core platform teams by designing an end-to-end full-stack RBAC service (Angular route guards, Express middleware, Active Directory integration) with fail-closed error handling, cutting authorization turnaround from hours to minutes.
- Accelerated migration of 100+ ETL pipelines by co-engineering a config-driven serverless data platform (AWS Glue, PySpark) with automated event ingestion, cutting dev effort by 80%, reducing data latency by 50%, and contributing to a patent filing.

Meta | Software Engineer Intern May 2022 - Aug. 2022

- Prevented unauthorized data access and safety policy bypasses by engineering 2 distributed auditing pipelines in Python and Presto, executing automated compliance and anomaly detection across petabyte-scale internal data warehouses.
- Architected full-stack privacy governance tooling (React, GraphQL, PHP/Hack) providing real-time user-access monitoring and automated policy remediation workflows for security and compliance teams.

CIDAR Lab | Software Engineer Intern June 2021 - Sept. 2021

- Integrated A* and Hightower pathfinding algorithms into open-source CAD tool 3DuF for microfluidic component routing, while refactoring 20% of the codebase from legacy JavaScript to TypeScript and Vue.js.

SKILLS & CERTIFICATIONS

Languages & Frameworks: Python, TypeScript, JavaScript, C/C++, SQL, React, Angular, Vue.js, Node.js, GraphQL

Cloud, AI & Infrastructure: AWS (Lambda, SQS, Glue), Claude Code, PySpark, Docker, PKI & Cryptography

Certifications: AWS Certified Developer, CompTIA Security+

EDUCATION

Georgia Institute of Technology | Atlanta, GA Aug. 2025 - *Expected Aug. 2027*

- Master of Science in Computer Science (*Human-Centered AI*)

Boston University | Boston, MA Sept. 2020 - May 2023

- Bachelor of Arts in Computer Science (*Cum Laude*)

LEADERSHIP & COMMUNITY

Capital One | Engineering & Cybersecurity Mentor Jan. 2026 - *Present*

- Facilitated weekly mentorship sessions for 5 incoming Discover engineers during post-acquisition integration, championing an inclusive engineering culture while accelerating technical onboarding and career development.

Boston University | Teaching Assistant Aug. 2021 - May 2023

- Collaborated with professors to develop curriculum and facilitated 12 discussion sections for undergraduate courses, including Introduction to Analysis of Algorithms (CS 330) and Computer Systems (CS 210).